DATA ANALYSIS PROJECT

HR Analytics Project

Dataset Overview and Analysis Plan

1. Dataset Inspection and Overview

Table	Contents
Employee	Core employee demographics, job roles, departments, compensation details, hire dates, tenure metrics, and attrition status.
PerformanceRating	Evaluation metrics including review dates, environment and job satisfaction scores, work-life balance, training opportunities, and performance ratings from both self and managers.
Dimension tables (EducationLevel, RatingLevel, SatisfiedLevel)	Standardized lookup tables that map categorical code numbers to descriptive text values.

2. Proposed Data Analysis Plan

The project will follow a structured workflow from data preparation through to reporting.

#	Phase	Description
1	**Data Cleaning and Preparation**	Inspect data types, handle missing values, resolve date formatting artifacts (such as Excel display errors), and merge dimension tables with core data.
2	**Exploratory Data Analysis (EDA)**	Analyze statistical distributions of employee salaries, ages, tenure, and overall attrition rates across various departments.
3	**Relationship and Attrition Analysis**	Identify key drivers behind employee turnover and examine correlations between workplace satisfaction scores and performance ratings.
4	**Business Insights and Recommendations**	Formulate actionable HR strategies to enhance employee retention, satisfaction, and performance management based on empirical findings.
5	**Python Code Development**	Write clear, executable Pandas, NumPy, Matplotlib, and Seaborn code formatted for Jupyter Notebook.
6	**PDF Report Compilation**	Generate a polished professional report summarizing the methodology, visual charts, and final conclusions.